



Comparisons

1. Class Recording Permission

One of our participants is affected by the current shutdown and cannot attend sessions. To support him, I'd like to record today's class.

If you prefer not to appear in the video, you're welcome to turn off your camera.

2. Speaking During Class

Just like last time, since there are many of us and slides take up my screen space, I might not see everyone.

If you'd like to contribute or ask a question, please feel free to unmute yourself and start speaking.

How People Process Information from Charts and Tables

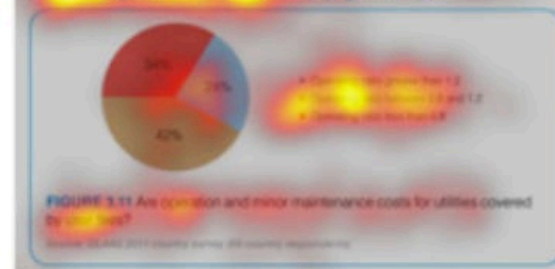
Summary data from respondent countries indicate that median government expenditure on sanitation and drinking-water is one sixth of that for health and one sixth of that for education (Table 3.4).

TABLE 3.4 Government expenditure on health, education and WASH

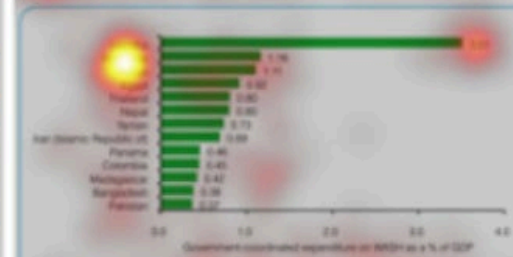
Country	Government expenditure on health (% of GDP)	Government expenditure on education (% of GDP)	Government expenditure on sanitation and drinking-water (% of GDP)
Armenia	3.1	3.4	0.4
Burkina Faso	5.4	4.7	0.3
Chad	1.9	3.6	0.8
Guatemala	4.6	—	1.2
India	1.3	—	0.2
Kenya (national government only)	2.3	4.7	0.7
Kenya (total)	3.1	6.9	1.1
Lesotho	8.4	—	0.5
Madagascar	2.7	3.2	0.4
Mali	1.7	4.7	0.8
Pakistan	9.1	3.9	0.5
Thailand	3.7	4.1	0.8
Togo	1.8	—	0.7
Tunisia	1.1	3.2	0.2
Yemen	9.4	14.0	0.5
Zambia	2.3	5.5	0.7

2010 data.
2009 data.
Not all countries reported contributions from regional and local governments.
Source: 2012 SLARS country survey; UNICEF (2012); WHO (2012); World Bank (2012).

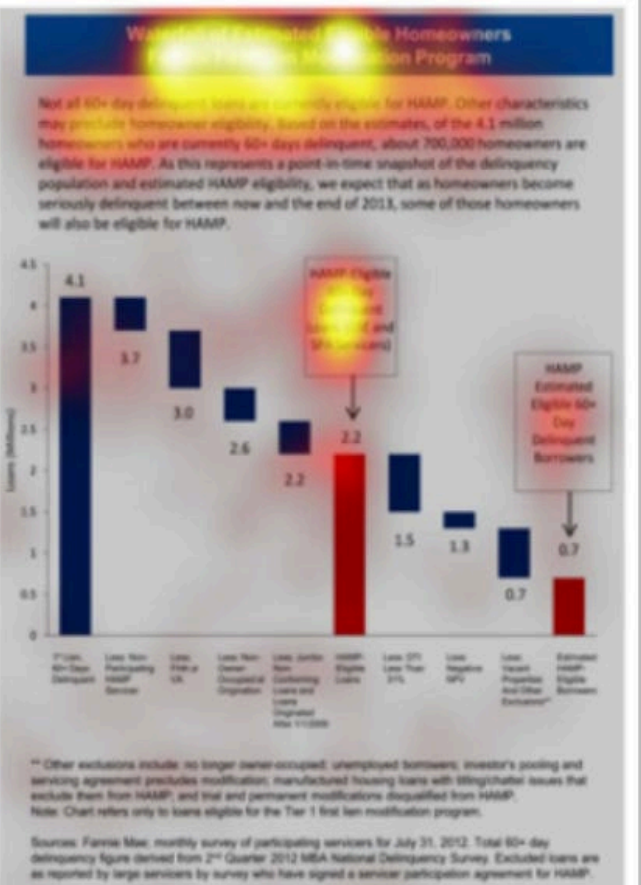
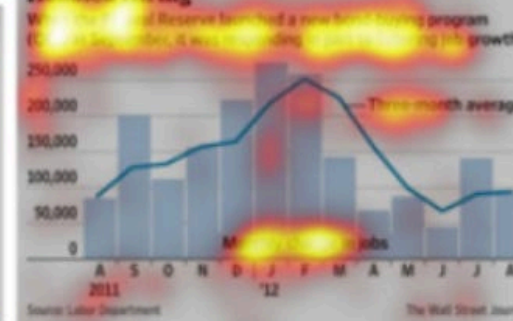
One third of countries indicate that revenues cover less than 80% of operating and minor maintenance costs for utilities covered by water users.



Government expenditure (from taxes and transfers) on sanitation and drinking-water ranged from 0.37% to 3.5% of GDP (Figure 3.2).



Hobbled Hiring



How People Process Information from Charts and Tables

Headlines capture attention.

They perfectly illustrate the “layered cake” scanning pattern: people fixate on titles first when viewing information.

People remember visualizations by their titles.

That’s one more reason not to label a chart with something generic like “Chart” or “Figure 1.”

Key insights are recalled through the title.

If the headline conveys the main idea of the visualization, people remember it easily; if it’s too vague, the idea fades.

People rarely notice bias in titles.

Even when a headline contradicts the visualization, most viewers don’t notice the inconsistency

- Why does Tufte call “**compared to what?**” the fundamental analytical act?
- Can you think of an example from your own work or research where forgetting to ask “compared to what?” led to misleading conclusions? or just suboptimal conclusions?

Think about the pie charts you see: do they help tell the story of the data, or not really?

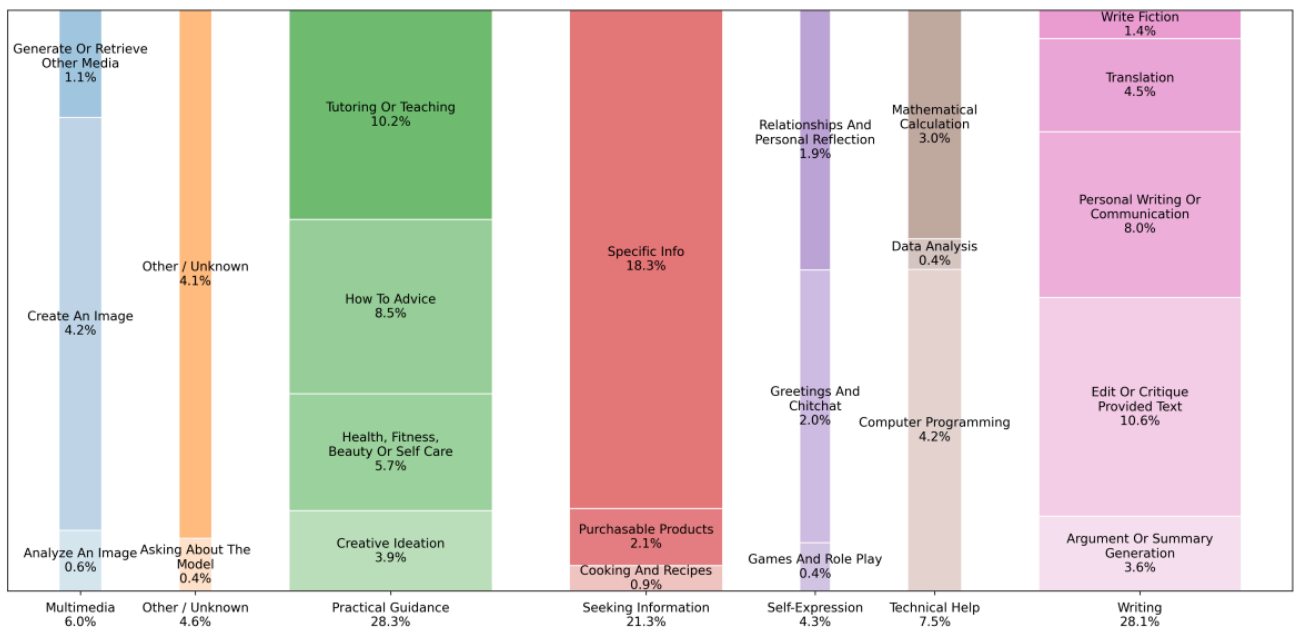
LABOUR FORCE PARTICIPATION RATES OF WOMEN AND MEN (AGED 25 TO 54), BY REGION AND HOUSEHOLD TYPE (PERCENTAGE)



Source: ILO and UN Women calculations from ILOSTAT database.
Note: Oceania excludes Australia and New Zealand. For the SDG region of Eastern and South-Eastern Asia, data are shown only for the subregion of South-Eastern Asia due to insufficient country coverage.

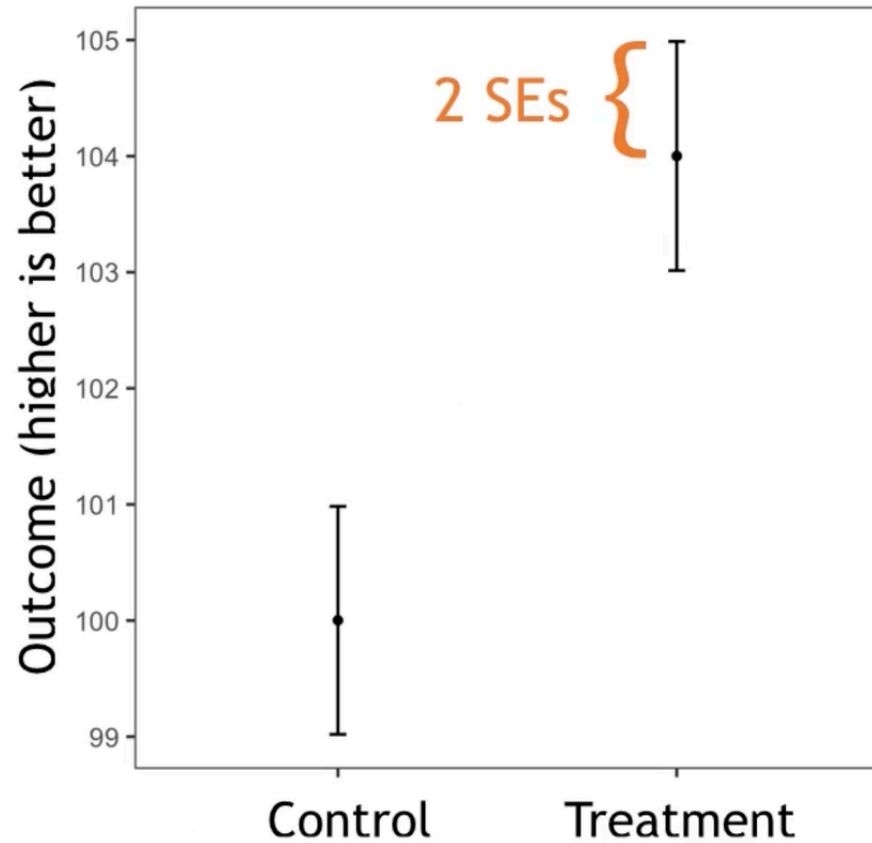


Male Female

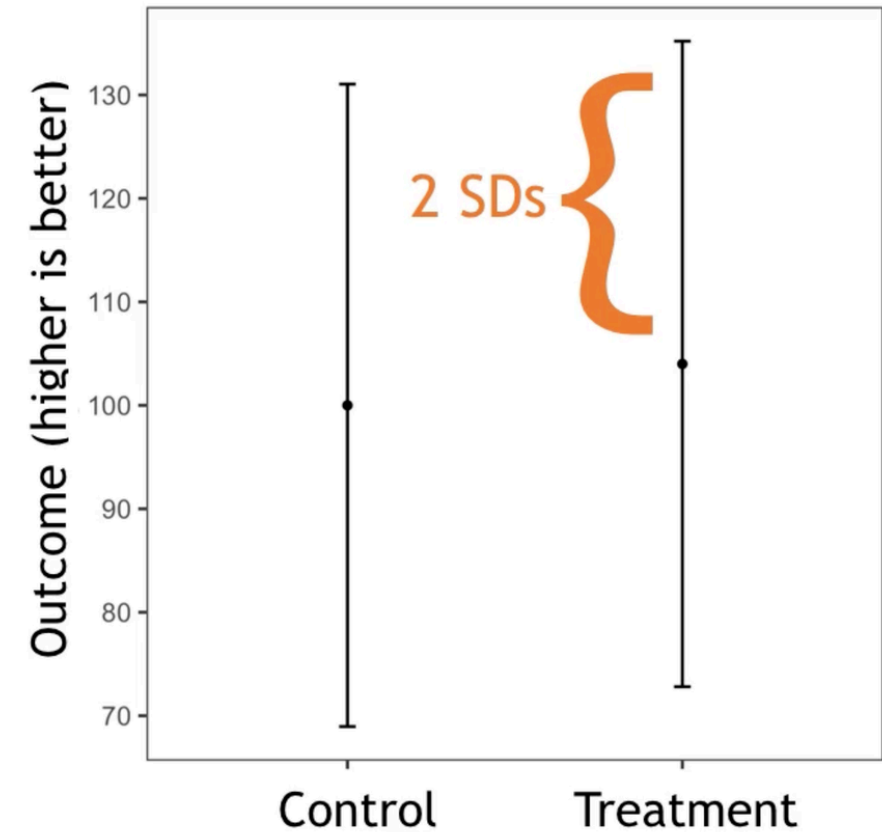


Comparing to what?

Figure 9: Breakdown of granular conversation topic shares within the coarse mapping defined in Table 3. The underlying classifier prompt is available in Appendix A. Each bin reports a percentage of the total population. Shares are calculated from a sample of approximately 1.1 million sampled conversations from May 15, 2024 through June 26, 2025. Observations are reweighted to reflect total message volumes on a given day. Sampling details available in Section 3.



Standard error (SE) bars show uncertainty in estimating the mean



Standard deviation (SD) bars show variation in outcomes around the mean

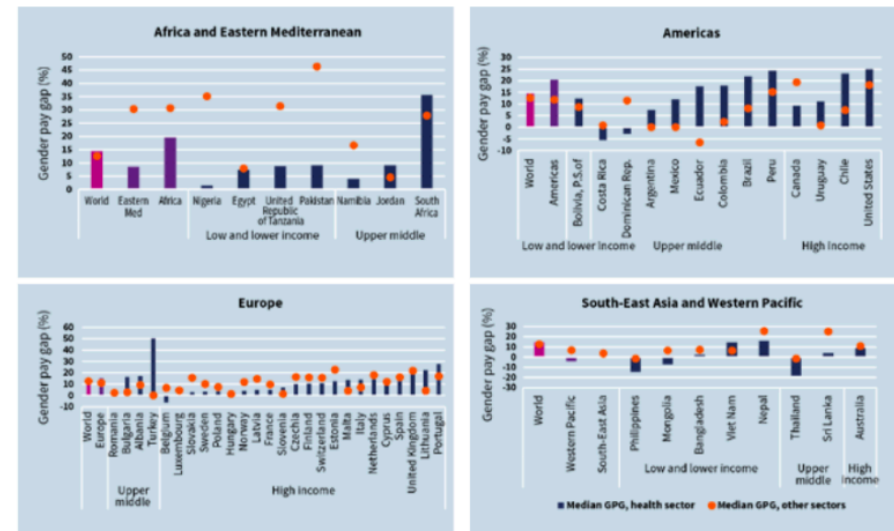
How Visualizing Inferential Uncertainty Can Mislead Readers About Treatment Effects in Scientific Results

Raw gender pay gaps using hourly wages

MEAN GENDER PAY GAP



MEDIAN GENDER PAY GAP



Source: ILO, WHO estimates based on survey data provided by national sources (see Annex 1).

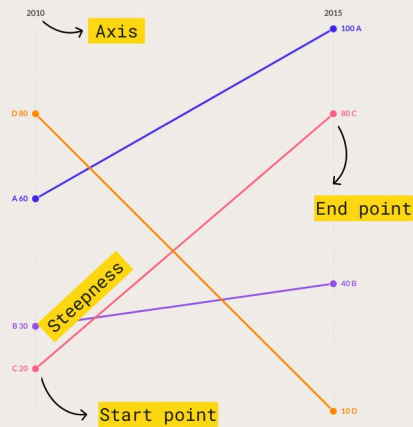
This graph shows both mean and median gender pay gap for health industry vs other industries

Figure 2: Raw gender pay gaps using hourly wages

- What chart type can we use to compare before and after
- Have you ever used it, and if yes for what?

Basics of a slope

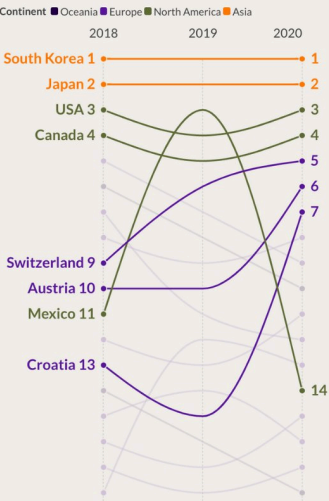
Slope charts are made up of **axes and lines** which represent the different **series**. The **thickness** of the lines can be linked to **values**. The **direction** of the line and the **steepness** of the slope convey the main meaning.



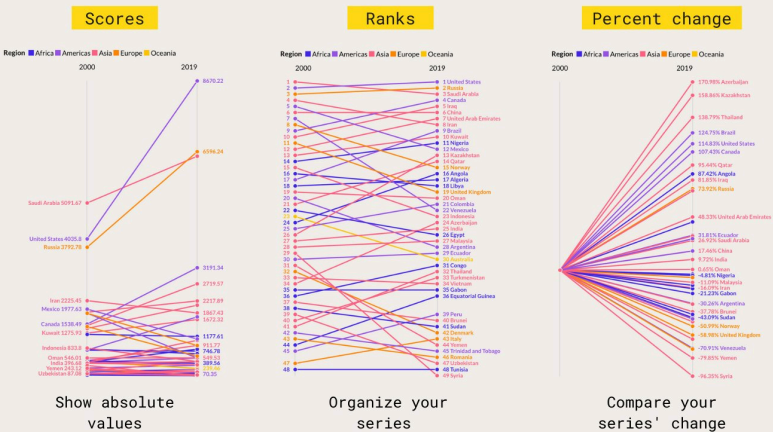
Slopes

What are slopes good for?

- Identifying and **showing change or variance** in a simplified manner
- Quickly **spotting general trends**
- Comparing** the data between two years



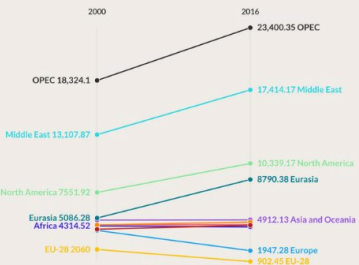
Three modes



Slope vs. Line charts

These are sister charts, but have different use cases.

Slope chart

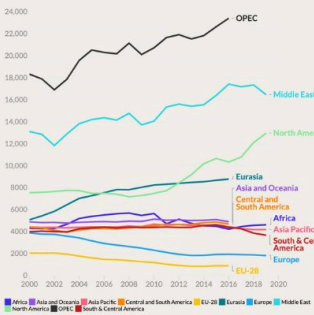


Similarities

- Both can represent time series.
- Every line is a series.
- Time is on X axis and values on Y axis.

Differences

- Slopes work better with less series than line charts.
- Slopes can show percentage change and ranks.
- Slopes don't need a zero base.



Line chart

- What chart type can we use to compare to a standard (explicit or implicit)
- Should we make out implicit standard explicit?

- Sparklines: who wants to explain?

28 day summary with change over previous period

Tweets

66 ↓21.4%



Tweet impressions

1.4M ↑5.1%



Profile visits

21.9K ↓9.1%



Mentions

447 ↓9.3%

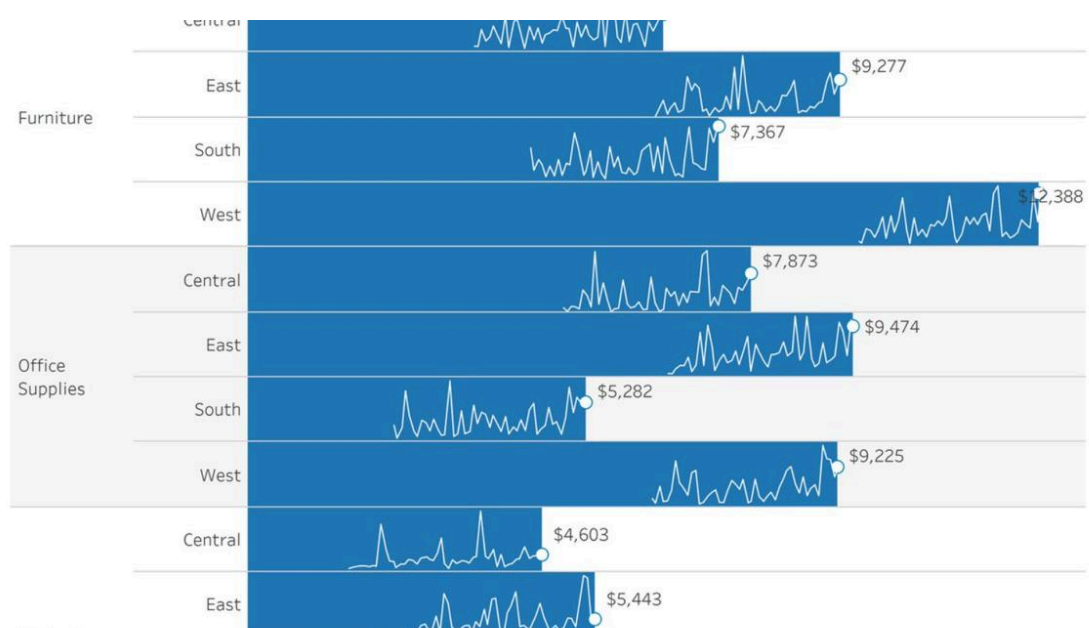


Followers

56.8K ↑1,143

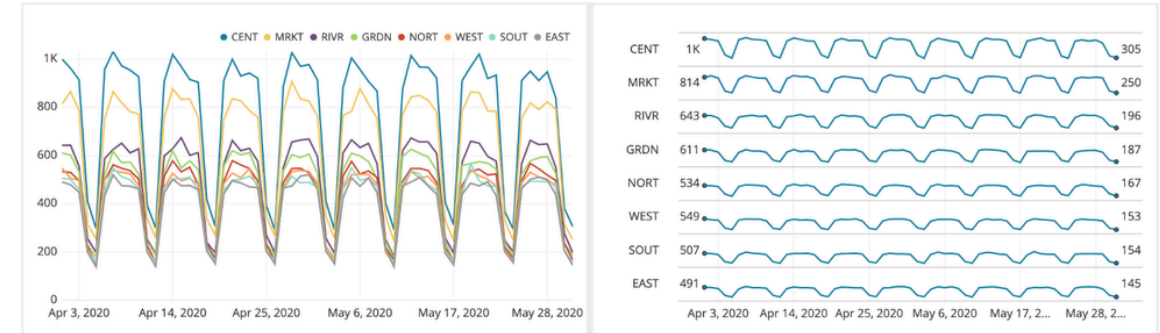


October 2015 • 3 days so far...

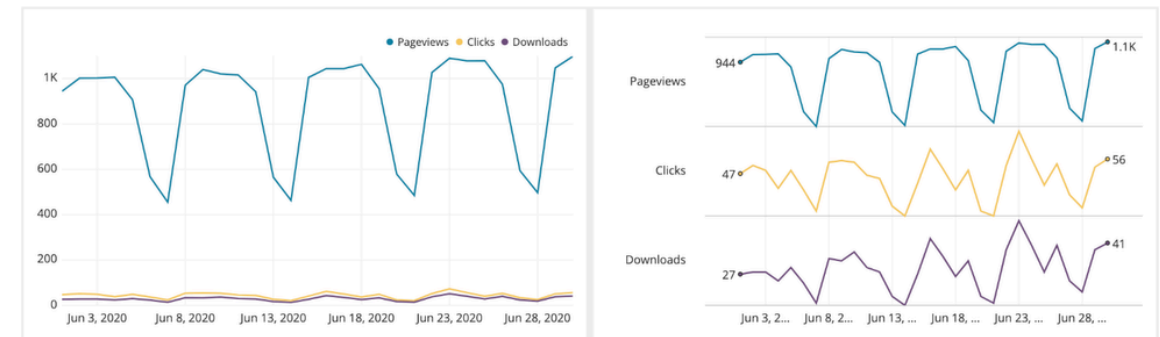


When you should use a sparkline

Like the line chart, a sparkline is great for showing how values change over time. The sparkline can serve as a useful alternative to the line chart when multiple lines need to be plotted.



On one extreme, the sparkline is a good choice when there is a lot of overlap between lines that cause them to be difficult to distinguish from one another.



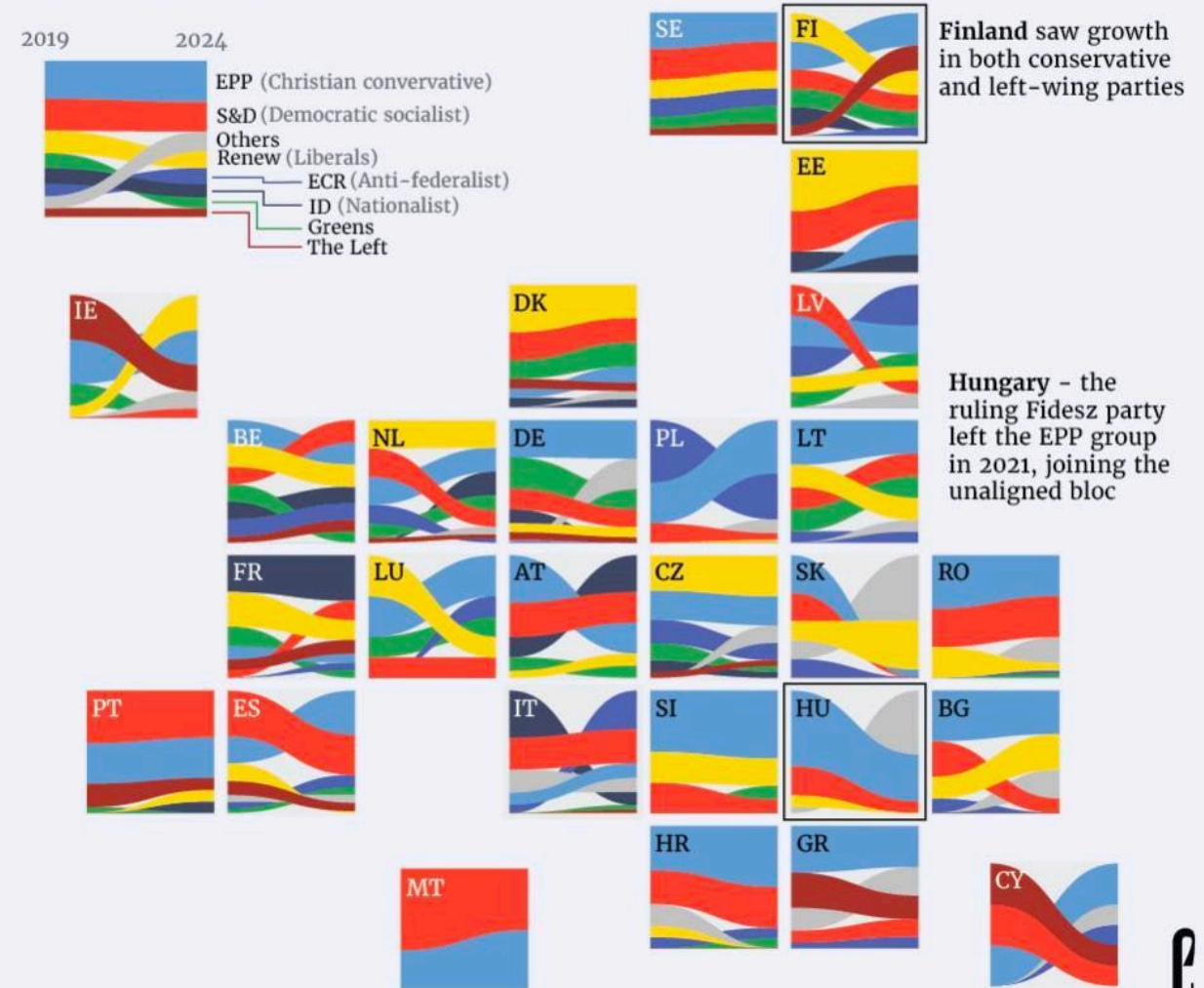
At the other extreme, sparklines are a good choice when there is no overlap between lines due to them covering broadly different ranges. This often pops up when you have multiple metrics that you want to track at the same time. Plotting all of the lines on a



- Small multiples

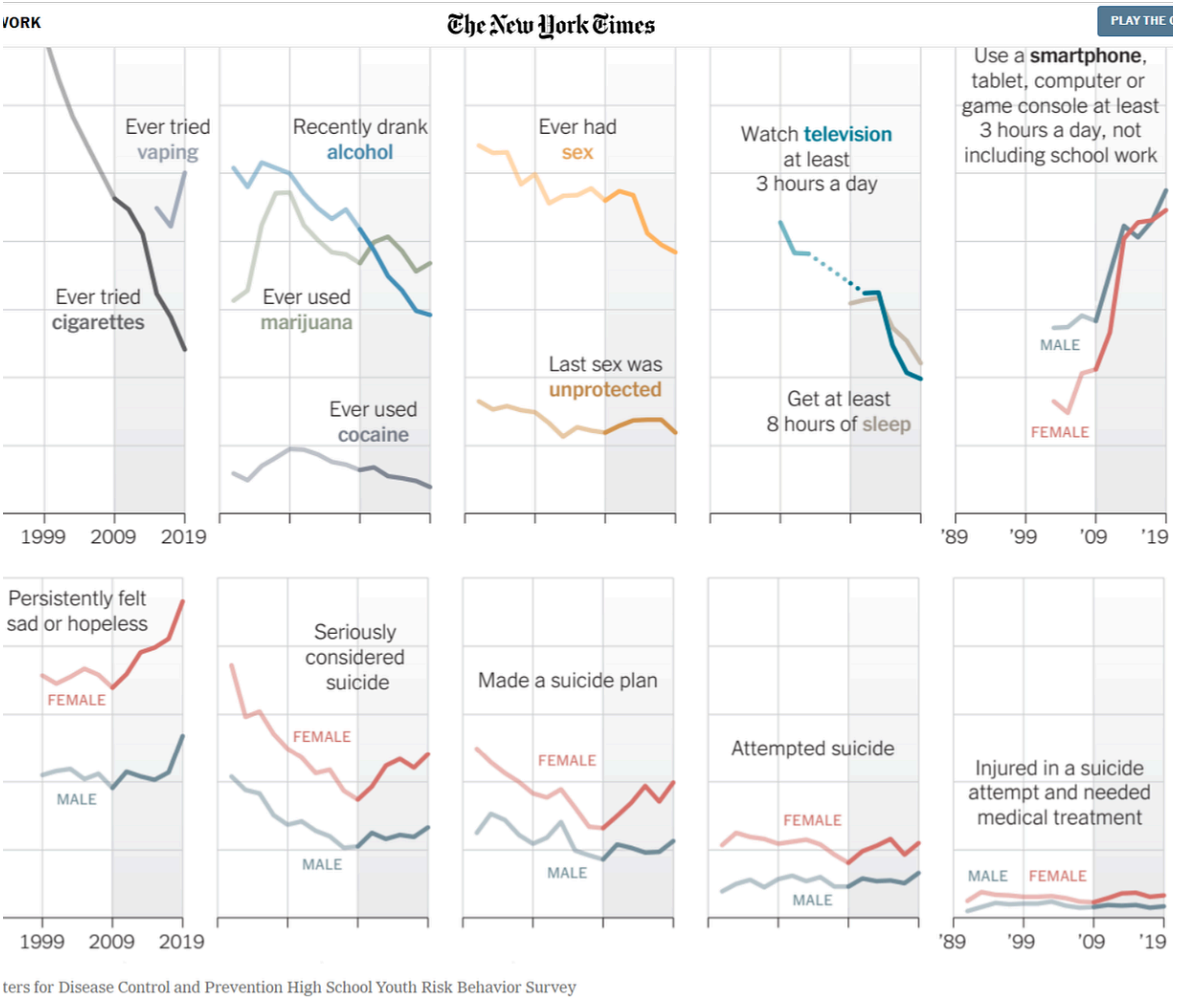
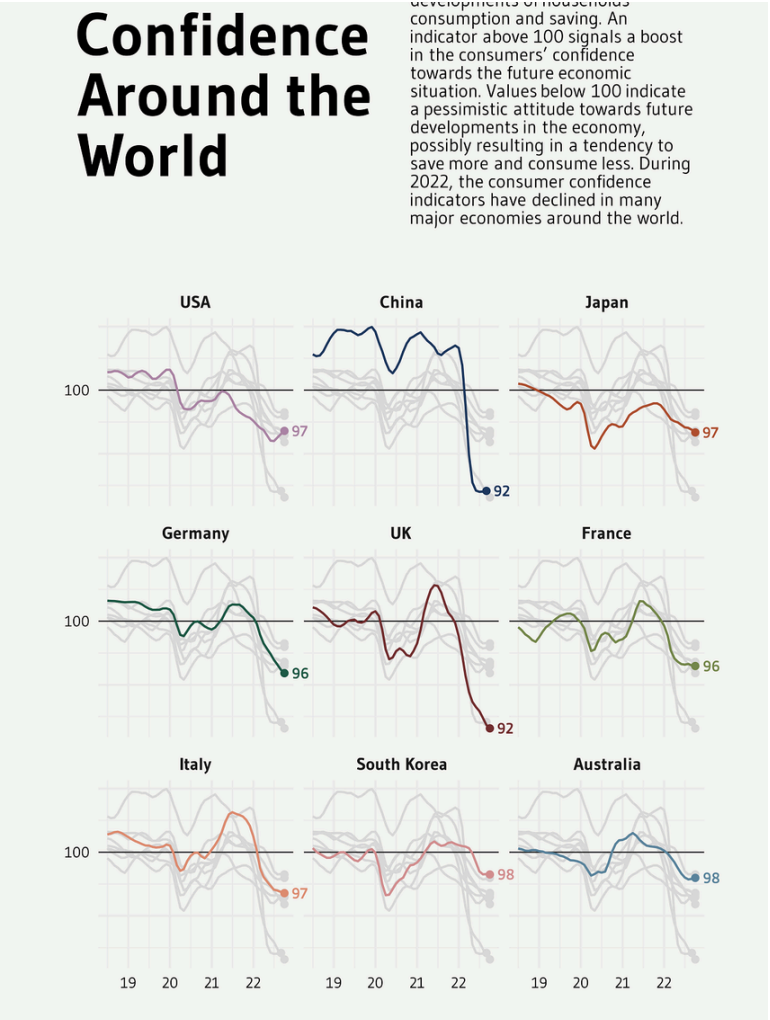
EU election overview

Change in vote share of main parties in EU elections



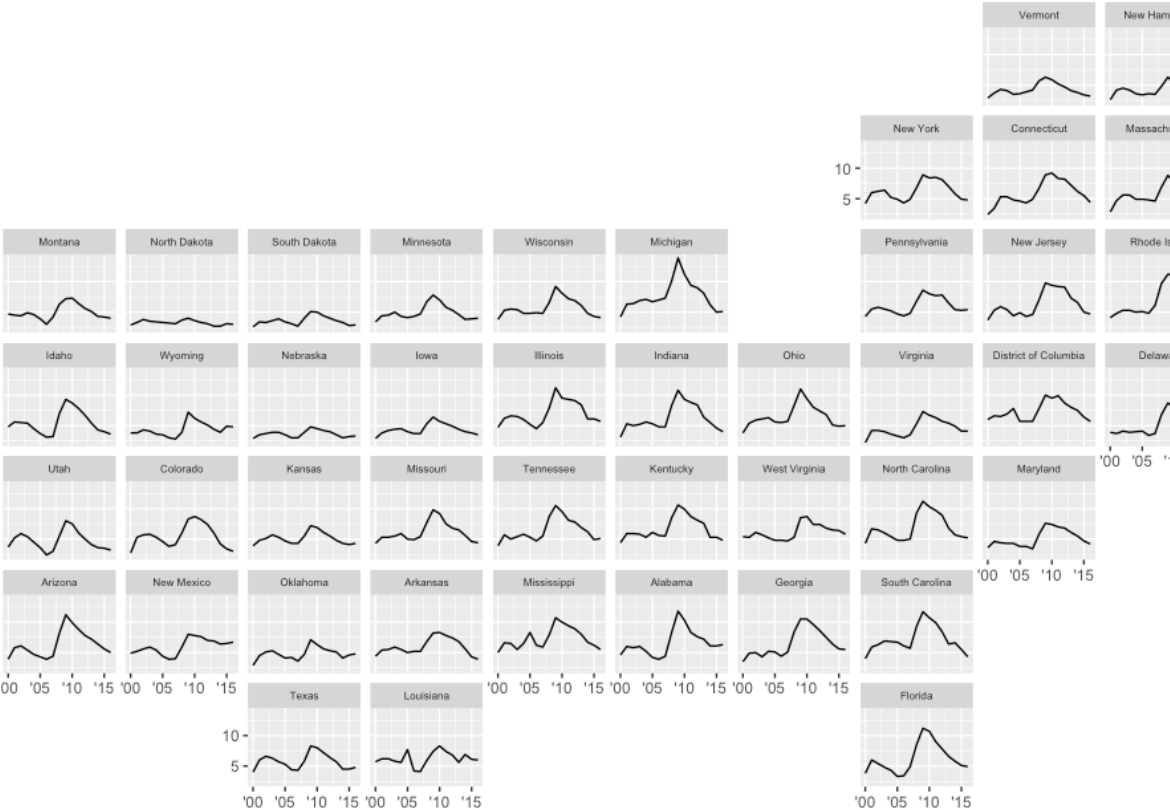
Source: European Union (2024)

How much is too much?



How much is too much?

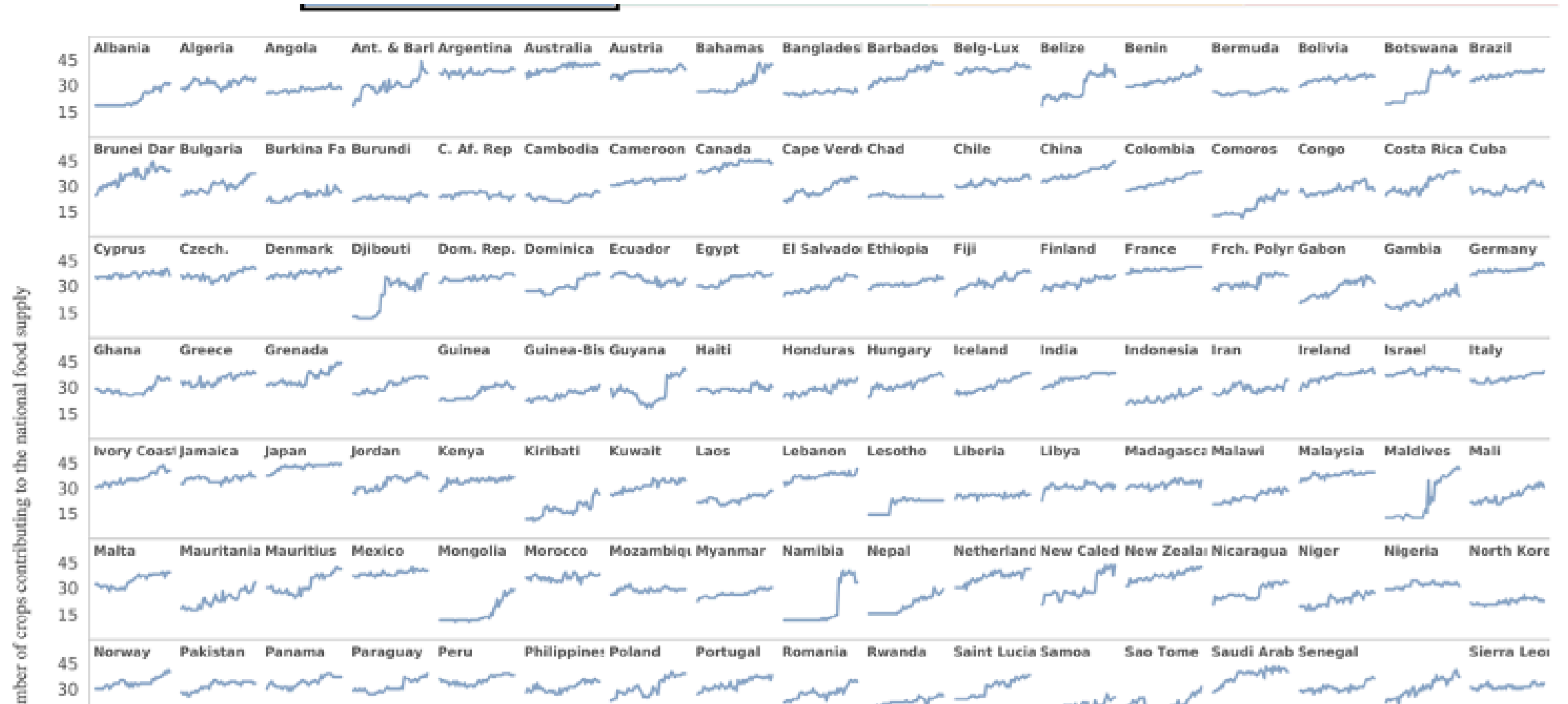
Adjusted US Unemployment Rate 2000-2016



Year



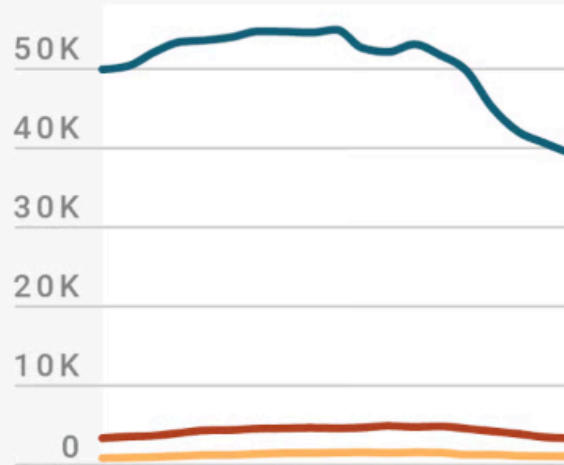
How much is too much?



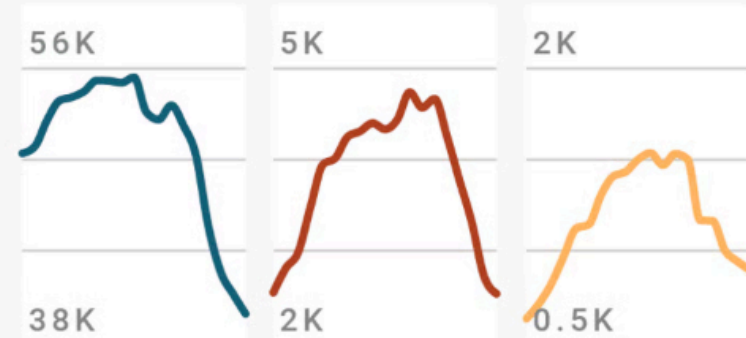
Some ideas regarding when to use small multiples

- when the number of groups is large (>5) for which traditional visualizations like bar charts and line charts may look messy and the storytelling is ineffective.
- Use small multiple line charts to untangle overlapping lines
- Normal line charts are better if you want readers to compare lines with each other at specific points in time
- You can make individual cells very small, our eye can spot trends

Different scales?



GOOD

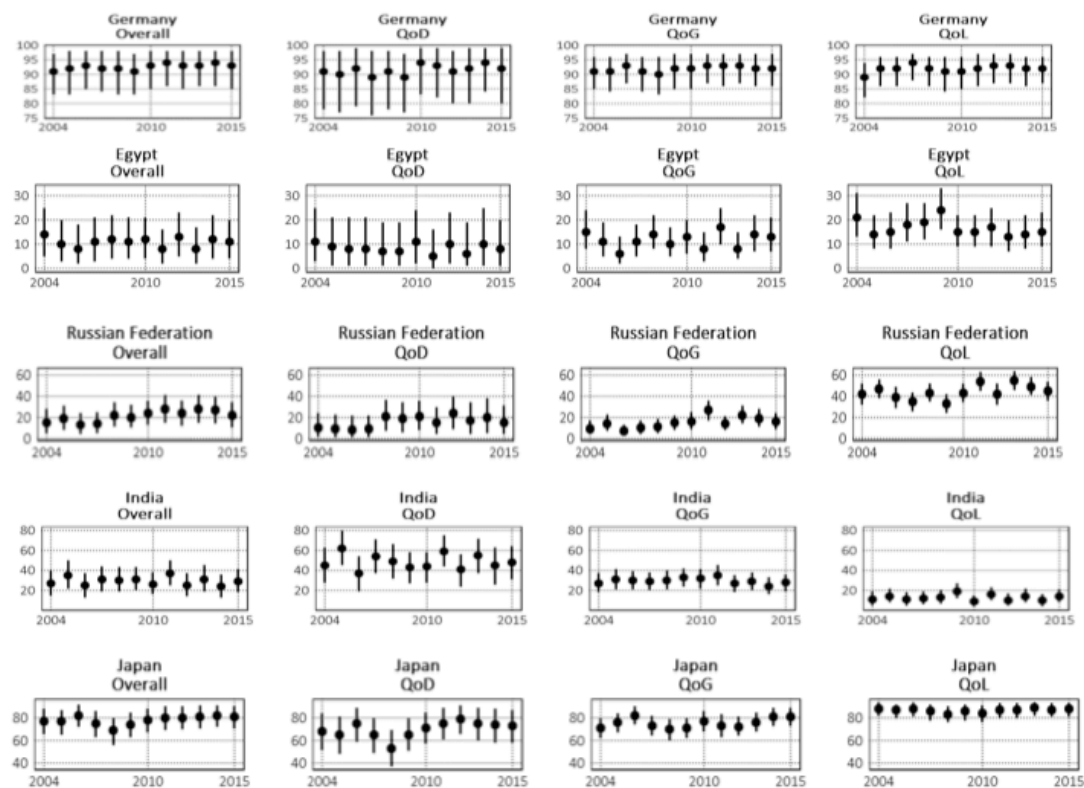


ALSO GOOD

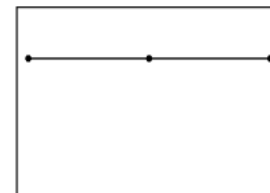
How to sort

- Alphabetically
(Often used as the default sorting method, useful when users know what they're looking for, does **not** help reveal patterns or insights)
You can always default to it when there is no obvious logic in your data
- Numerical Value
(select 1 value, the first one, the last one, overall change of positions etc)
- Aggregated Value
(Can sort by average, maximum, or minimum values)
- Pattern/Structure
(could be geographical, could be pattern in the data etc)

Finding patterns

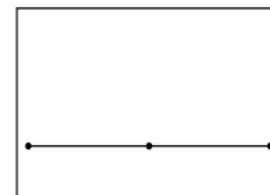


similar high levels of
Quality of Democracy,
Quality of Government and
Quality of Life



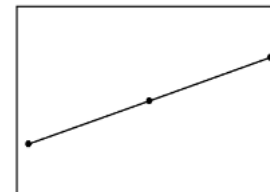
Australia, Canada,
Germany,
Netherlands, Norway

similar low levels of
Quality of Democracy,
Quality of Government and
Quality of Life



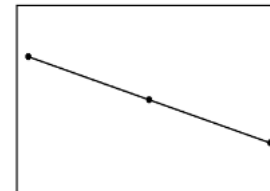
Argentina, Egypt,
Georgia, Nigeria,
Philippines

Quality of Life is better
than Quality of Democracy
and Quality of Government



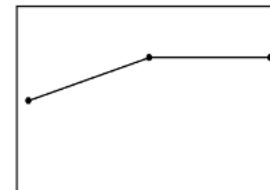
China, Kazakhstan,
Russian Federation

Quality of Life is less
than Quality of Democracy
and Quality of Government



Brazil, Colombia,
India, Italy

Quality of Government and
Quality of Life is better
than Quality of Democracy



Japan, Korea